

On the Theory of Modules

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The following investigations concern those modules whose introduction into number theory is due to *Dedekind*. In the eleventh supplement to *Dirichlet's* lectures edited by him, the theory of modules is presented in its basic features. In addition, two further papers must be considered here: *Frobenius*, Theory of linear forms with integral coefficients; *Frobenius* and *Stickelberger*, On groups of interchangeable elements (Journal für die reine und angewandte Mathematik, vols. 86, 88). In the latter, the connection between modules and groups of interchangeable elements is made clear, and after this it is easy to transfer results obtained in one theory to the other. I shall therefore confine myself to the treatment of modules, although it would be natural to include group-theoretic questions.

I first state the most important relevant theorems of *Dedekind* and *Frobenius*, in the form and with the notation which I shall use throughout what follows.

§ 1. Rectangular systems; their division into classes.

1. Rectangular systems.

In what follows, capital Latin letters denote systems of rational integers arranged in rows and columns.¹ On such systems the operations addition, subtraction, and multiplication are to be applied in the usual way. Addition and subtraction are possible when the rectangular systems contain the same number of rows and the same number of columns; multiplication is possible when the number of columns of each factor is equal to the number of rows of the following factor, and the product then has as many rows as its first factor and as many columns as its last factor.

The number of columns of a rectangular system A is called its degree. If this degree is n and $0 < k \leq n$, then the greatest common divisor of the determinants of degree k in A is called “the k th determinant-divisor of A ”. It is also to be noted that the greatest common divisor of given numbers is taken positive and is to be put equal to 0 only when all the numbers are zero. If the number r of rows of A is $< n$, then nevertheless n determinant-divisors are to be counted, by stipulating that for $k > r$ the k th determinant-divisor is to be put equal to 0. The number of non-zero determinant-divisors of a rectangle is called its “rank”.

If the system A is square, it is called a diagonal system when all elements outside the diagonal vanish. If $a_1, a_2, \dots, a_n$ are the diagonal elements of such a system, it will also be denoted by $\text{Dg}(a_1 \mid a_2 \mid \dots \mid a_n)$. This diagonal system is called a “principal system” when none of the elements a is negative and each is a divisor of the following one. A square system A is called unimodular when its “norm” ($\text{Nm } A$), that is, the absolute value of its determinant, is equal to 1. The letter E will always denote only unimodular systems; the symbol E_+ will denote those of determinant $+1$. If E^0 is the diagonal system of degree n all of whose diagonal elements are equal to 1, then for every square system A of degree n one has $AE^0 = A$, $E^0A = A$, and to every unimodular system E there belongs a second one E^{-1} for which

$$EE^{-1} = E^{-1}E = E^0$$

holds.

2. Division of square systems into classes.

We shall first restrict ourselves to considering square systems.

¹The assumption that the elements are rational integers will be maintained throughout.

Definition 1. If A and B are square systems of the same degree n , and if A can be represented as a product in which B occurs as a factor, then we say: “The class of the system A ($\text{Cl } A$) is divisible by the class of the system B .”

If $\text{Cl } A$ is divisible by $\text{Cl } B$ and $\text{Cl } B$ is divisible by $\text{Cl } C$, then, as is immediately seen, $\text{Cl } A$ is also divisible by $\text{Cl } C$.

Definition 2. Two classes are called equal when each is divisible by the other.

The admissibility of these definitions is easily verified. The square systems are thereby divided into classes; each class is completely determined by a single system contained in it.

The problems now arise: given two square systems A and B , decide whether $\text{Cl } A$ is divisible by $\text{Cl } B$, or whether the systems A and B belong to the same class. These questions are decided by two theorems, probably first developed by *Frobenius*, which in what follows will be called Fundamental Theorems I and II.

Fundamental Theorem I. In order that two square systems A and B belong to the same class, it is necessary and sufficient that they possess the same determinant-divisors. If this requirement is fulfilled, A can always be represented in the form

$$A = EBE'$$

and, if in addition the determinants of A and B agree also in sign, also in the form

$$A = E_+BE'_+.$$

To this theorem there is joined a second one, which at the same time reveals an important property of the determinant-divisors.

Ia. Every class contains one and only one principal system.

Let $\text{Dg}(e_1 \mid e_2 \mid \cdots \mid e_n)$ be a principal system, and let $d_1, d_2, \dots, d_n$ be its determinant-divisors. Then $d_k = e_1 e_2 \cdots e_k$. Hence, if $d_1, d_2, \dots, d_n$ are the determinant-divisors of any square system and the first r of them are non-zero, then

$$e_1 = d_1, \quad e_2 = \frac{d_2}{d_1}, \quad \dots, \quad e_r = \frac{d_r}{d_{r-1}}, \quad e_{r+1} = 0, \dots, e_n = 0$$

are the elements of the principal system of its class; hence, among the quotients

$$\frac{d_1}{1}, \quad \frac{d_2}{d_1}, \quad \dots, \quad \frac{d_r}{d_{r-1}},$$

each is a divisor of the following one.

The diagonal elements of the principal system are called the “elementary divisors” or “invariants” of its class, or also of every system of this class. In what follows classes are denoted by capital Greek letters, or by giving their invariants; thus, if $a_1, a_2, \dots, a_n$ are the invariants of $\text{Cl } A = A$, we write

$$A = \text{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n).$$

All systems of a class A agree in degree, rank, norm, and determinant-divisors, and consequently one can speak of the degree, rank, and so on of the class A .

Fundamental Theorem II. The class A is divisible by the class B if and only if each invariant of A is divisible by the corresponding invariant of B .

To simplify expression, we now introduce a few elementary notations. The greatest common divisor of s numbers $a_1, a_2, \dots, a_s$ will be denoted by

$$(a_1, a_2, \dots, a_s),$$

and their least common multiple by

$$[a_1, a_2, \dots, a_s].$$

The latter is to be put equal to 0 as soon as one of the numbers a vanishes; otherwise it is to be taken positive. Now let $A_1, A_2, \dots, A_s$ be classes of degree n , let

$$A_i = \text{Cl}(a_{i1} \mid a_{i2} \mid \dots \mid a_{in}) \quad (i = 1, \dots, s),$$

and put

$$(a_{1k}, a_{2k}, \dots, a_{sk}) = d_k, \quad [a_{1k}, a_{2k}, \dots, a_{sk}] = c_k \quad (k = 1, \dots, n).$$

Then $d_1, d_2, \dots, d_n$ form the system of invariants of a certain class of degree n , and the same is true of the numbers $c_1, c_2, \dots, c_n$. The first of these classes is denoted by

$$(A_1, A_2, \dots, A_s),$$

the second by

$$[A_1, A_2, \dots, A_s].$$

Thus $(A_1, A_2, \dots, A_s)$ is the greatest common divisor, and $[A_1, A_2, \dots, A_s]$ the least common multiple, of the classes $A_1, A_2, \dots, A_s$. That is, the divisors common to all the classes $A_1, \dots, A_s$ are exactly the divisors of $(A_1, \dots, A_s)$, and the multiples common to all the classes $A_1, A_2, \dots, A_s$ are exactly the multiples of $[A_1, \dots, A_s]$. The class of degree n whose invariants are all equal to 1, and which is therefore a divisor of every class of degree n , will be denoted by $E^{(n)}$ (or simply by E when the degree is clear from the context). Two classes A, B are called relatively prime when $(A, B) = E$.

3. Classes of positive norm.

Let $A = \text{Cl}(a_1 \mid a_2 \mid \dots \mid a_n)$ be a class of positive norm, and let $\text{Nm } A = a$ be the product of the relatively prime numbers a' and a'' . Then A has only one divisor A' of norm a' and only one divisor A'' of norm a'' . These classes

$$A' = \text{Cl}((a_1, a') \mid (a_2, a') \mid \dots \mid (a_n, a')),$$

$$A'' = \text{Cl}((a_1, a'') \mid (a_2, a'') \mid \dots \mid (a_n, a''))$$

are called “components” of A ; A is “decomposable” into the components A' and A'' . The characteristic relations for this decomposability are

$$(A', A'') = E, \quad [A', A''] = A.$$

The decomposition of A can be continued until one reaches primary classes, i.e. classes whose norm is a power of a prime. In general, if m is a positive integer and p a prime, let m_p denote the highest power of p dividing m ; correspondingly put

$$A_p = \text{Cl}((a_1)_p \mid (a_2)_p \mid \dots \mid (a_n)_p).$$

Then A_p is the primary component of A corresponding to the prime p . For primes p which do not divide $\text{Nm } A$, one has $A_p = E$. Every class A of positive norm can be decomposed into primary components in one way only, and is the least common multiple of these components. If A is divisible by B , then for every prime p , A_p is divisible by B_p ; and if for every prime p , A_p is divisible by B_p , then A is divisible by B . If $A_1, A_2, \dots, A_s, \Delta, \Gamma$ are classes of positive norm, then the relations

$$(A_1, A_2, \dots, A_s) = \Delta, \quad [A_1, A_2, \dots, A_s] = \Gamma$$

are respectively equivalent to the systems of relations

$$((A_1)_p, (A_2)_p, \dots, (A_s)_p) = \Delta_p, \quad [(A_1)_p, (A_2)_p, \dots, (A_s)_p] = \Gamma_p.$$

4. Division of rectangular systems into classes.

In order now to include all rectangular systems among the classes, stipulate that every rectangular system A belongs to the class which possesses the same determinant-divisors as A . If the invariants of the class $\text{Cl } A$ are also called the invariants of the system A , then the following theorem holds:

If A and B are rectangular systems, then in order that an equation of the form

$$A = P \cdot B \cdot Q$$

be possible, it is necessary and sufficient that $\text{Rg } A \leq \text{Rg } B$ and that each positive invariant of A be divisible by the corresponding invariant of B .

If A and B have the same degree, this condition amounts to $\text{Cl } A$ being divisible by $\text{Cl } B$. If the rectangles A and B have the same degree (n) and the same number of rows (r), and if $\text{Cl } A$ is divisible by $\text{Cl } B$, then in an equation of the form $A = P \cdot B \cdot Q$ the systems P and Q are square systems; and if $\text{Cl } A = \text{Cl } B$, it can always be arranged that P and Q are unimodular.

§ 2. Modules.

1. Modules.

According to *Dedekind*, a system of numbers is called a “module” if the difference of any two numbers of the system again belongs to the system. The module $\mathfrak{a}$ is said to be divisible by the module $\mathfrak{b}$ (a multiple of $\mathfrak{b}$, and $\mathfrak{b}$ a divisor of $\mathfrak{a}$) when every number of $\mathfrak{a}$ is also contained in $\mathfrak{b}$. If the numbers

$$\alpha_1, \alpha_2, \dots, \alpha_r$$

belong to the module $\mathfrak{a}$, then the same is true of every number of the form

$$\alpha = c_1\alpha_1 + c_2\alpha_2 + \dots + c_r\alpha_r, \tag{1}$$

where the coefficients c are rational integers. The module $\mathfrak{a}$ is called finite when one can take from it a finite number of numbers $\alpha_1, \alpha_2, \dots, \alpha_r$ such that every number of $\mathfrak{a}$ can be represented in the form (1). The numbers $\alpha_1, \alpha_2, \dots, \alpha_r$ are then said to form a basis of $\mathfrak{a}$, and this will be indicated by writing

$$\mathfrak{a} = \text{Md}(\alpha_1, \alpha_2, \dots, \alpha_r).$$

The basis is called reduced if the expression (1) can vanish only by all coefficients vanishing. Then every number α of $\mathfrak{a}$ can be brought into the form (1) in only one way; every basis of $\mathfrak{a}$ consists of at least r numbers; and the module $\mathfrak{a}$ is called r -component. Every multiple of an r -component module is at most r -component.

Suppose one has an n -component module

$$\mathfrak{e} = \text{Md}(\eta_1, \eta_2, \dots, \eta_n).$$

If one considers only numbers of this module and performs no operations on them other than addition and subtraction, the basis elements $\eta_1, \eta_2, \dots, \eta_n$ may be treated as indeterminate quantities. This amounts to considering, in place of the numbers contained in $\mathfrak{e}$

$$\eta = c_1\eta_1 + \dots + c_n\eta_n,$$

the systems of the n coefficients

$$c_1, c_2, \dots, c_n$$

by which the numbers η are assembled from the numbers $\eta_1, \eta_2, \dots, \eta_n$. Such a system of coefficients will always be thought of as arranged in a horizontal row. Doing this, while otherwise retaining the notation used so far, we shall understand by a module a totality of systems or “rows” σ which reproduce themselves under subtraction.

Now let

$$\mathfrak{a} = \text{Md}(\alpha_1, \alpha_2, \dots, \alpha_r)$$

be any module. If one forms the rectangle A whose rows are $\alpha_1, \alpha_2, \dots, \alpha_r$, then the module $\mathfrak{a}$ is completely determined by it. We call the rectangle A a basis of the module $\mathfrak{a}$ and write

$$\mathfrak{a} = \text{Md } A.$$

The necessary and sufficient condition that the module $\mathfrak{b} = \text{Md } B$ be divisible by the module $\mathfrak{a} = \text{Md } A$ is evidently the existence of an equation of the form

$$B = P \cdot A.$$

Taking account of the definitions and theorems given in § 1, it follows that:

If the rectangles $A_1, A_2, \dots$ are bases of one and the same module $\mathfrak{a}$, then they all belong to the same class. It is convenient to call this class the class of the module $\mathfrak{a}$, so that

$$\text{Cl } A_1 = \text{Cl } A_2 = \dots = \text{Cl } \mathfrak{a}.$$

By transferring to the modules belonging to a class the concepts previously introduced for the class itself, it is then clear what is to be understood by the degree, rank, norm of a module $\mathfrak{a}$ ($\text{Gd } \mathfrak{a}, \text{Rg } \mathfrak{a}, \text{Nm } \mathfrak{a}$), and by its determinant-divisors and invariants. One sees that the rank of a module is the same thing as what was earlier called its number of components, so that a module of rank r is r -component. Divisibility can of course occur only between modules of the same degree.

I now state several important theorems from the theory of modules, which will be used frequently later.

I. If $\text{Rg } \mathfrak{a} = r$, the module $\mathfrak{a}$ possesses a basis of k systems if and only if $k \geq r$. Since $\text{Gd } \mathfrak{a} \geq \text{Rg } \mathfrak{a}$, every module of degree n also possesses a basis of n systems, a square basis.

II. Every module $\mathfrak{a}$ possesses a square basis of the form AE , where A is the principal system of $\text{Cl } \mathfrak{a}$. The class $E^{(n)}$ contains only one module; this module is a divisor of every module of degree n , and will be denoted by $\mathfrak{e}^{(n)}$ (or $\mathfrak{e}$). Every unimodular system of degree n is a basis of $\mathfrak{e}^{(n)}$. Theorem II can therefore also be formulated as follows:

IIa. If $\text{Cl } \mathfrak{a} = A = \text{Cl}(a_1 \mid a_2 \mid \dots \mid a_n)$, one can give n rows $\alpha_1, \alpha_2, \dots, \alpha_n$ such that

$$\text{Md}(\alpha_1, \alpha_2, \dots, \alpha_n) = \mathfrak{e}, \quad \text{Md}(a_1\alpha_1, a_2\alpha_2, \dots, a_n\alpha_n) = \mathfrak{a}.$$

If $\text{Rg } \mathfrak{a} = r$, then also $\mathfrak{a} = \text{Md}(a_1\alpha_1, \dots, a_r\alpha_r)$.

III. If the module $\mathfrak{a}$ is a multiple of $\mathfrak{b}$, then $\text{Cl } \mathfrak{a}$ is a multiple of $\text{Cl } \mathfrak{b}$; and if the class A is a multiple of B , then every module of A has at least one divisor within B , and every module of B has at least one multiple within A .

IV. If $\mathfrak{a}$ is a multiple of $\mathfrak{b}$ and $\text{Cl } \mathfrak{a}$ is a divisor of $\text{Cl } \mathfrak{b}$, then $\mathfrak{a} = \mathfrak{b}$.

Let $\mathfrak{a}_1 = \text{Md } A_1, \mathfrak{a}_2 = \text{Md } A_2, \dots, \mathfrak{a}_s = \text{Md } A_s$ be modules of the same degree n . They possess a greatest common divisor, that is, there exists a definite module $\mathfrak{b}$ with the property that every divisor common to the modules $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$ is a divisor of $\mathfrak{b}$, and conversely. The rectangle whose rows are all the rows of the rectangles $A_1, A_2, \dots, A_s$ is a basis of this module. The modules $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$ also possess a least common multiple, that is, there exists a definite

module $\mathfrak{c}$ with the property that every common multiple of the modules $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$ is a multiple of $\mathfrak{c}$, and conversely. Simple considerations show how to find a basis of this module $\mathfrak{c}$ by a finite number of operations, even in the cases in which some of the modules $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$ have vanishing norm. We denote by

$$(\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s), \quad [\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s]^2$$

the greatest common divisor and the least common multiple of the modules $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$, respectively.

The fact that the module $\mathfrak{a}$ is a multiple of the module $\mathfrak{b}$ may be expressed by either of the equations

$$(\mathfrak{a}, \mathfrak{b}) = \mathfrak{b}, \quad [\mathfrak{a}, \mathfrak{b}] = \mathfrak{a},$$

but also, since $(\mathfrak{a}, \mathfrak{b})$ is in any case a divisor of $\mathfrak{b}$, by Theorem IV through the equation

$$\text{Cl}(\mathfrak{a}, \mathfrak{b}) = \text{Cl } \mathfrak{b}.$$

This gives a simple means of recognizing whether, for two modules $\mathfrak{a}, \mathfrak{b}$ each given by a basis, the first has the second as divisor. For then one also has a basis of $(\mathfrak{a}, \mathfrak{b})$, so one can immediately determine the class of the module $(\mathfrak{a}, \mathfrak{b})$ and has only to see whether it is identical with $\text{Cl } \mathfrak{b}$.

V. Between the norms of two modules $\mathfrak{a}, \mathfrak{b}$, their greatest common divisor, and their least common multiple, there is the relation

$$\text{Nm } \mathfrak{a} \text{ Nm } \mathfrak{b} = \text{Nm}(\mathfrak{a}, \mathfrak{b}) \text{ Nm}[\mathfrak{a}, \mathfrak{b}].$$

If A is the principal system of the class A , the module $\text{Md } A$ is called the “principal module” of the class A . In order that the class A be divisible by B , it is necessary and sufficient that the principal module of A be divisible by the principal module of B . The greatest common divisor (respectively, the least common multiple) of the principal modules of the classes $A_1, A_2, \dots, A_s$ is the principal module of the class $(A_1, A_2, \dots, A_s)$ (respectively, $[A_1, A_2, \dots, A_s]$). Two modules $\mathfrak{a}, \mathfrak{b}$ are called relatively prime when $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{c}$.

2. Unimodular transformations.

Let A and B be square systems of degree n , let E be a unimodular system of the same degree, and suppose there is an equation

$$A = CB. \tag{2}$$

Then it follows that

$$AE = CBE. \tag{3}$$

Conversely, (2) also follows from (3). This result can be stated as follows:

If $\text{Md } A$ is divisible by $\text{Md } B$, then $\text{Md } AE$ is divisible by $\text{Md } BE$, and conversely. Of the relations

$$\text{Md } A = \text{Md } B, \quad \text{Md } AE = \text{Md } BE, \tag{4}$$

each is therefore a consequence of the other. Thus, for every square basis A of a definite module $\mathfrak{a}$, AE represents the basis of a definite second module $\mathfrak{a}'$. We therefore say:

“The module $\mathfrak{a}$ is carried by the unimodular transformation $[E]$ into the module $\mathfrak{a}'$ ”

and write

$$\mathfrak{a}' = \mathfrak{a}E.$$

The inverse transformation $[E^{-1}]$ carries $\mathfrak{a}'$ back into $\mathfrak{a}$.

²In Dedekind's notation, $\mathfrak{a}_1 + \mathfrak{a}_2 + \dots + \mathfrak{a}_s$ and $\mathfrak{a}_1 - \mathfrak{a}_2 - \dots - \mathfrak{a}_s$, respectively.

From the equivalence of equations (2) and (3) it follows further that:

VI. If the module $\mathfrak{a}$ is divisible by the module $\mathfrak{b}$, then $\mathfrak{a}E$ is divisible by $\mathfrak{b}E$, and conversely.

A simple consideration then gives:

VII. The greatest common divisor (respectively, the least common multiple) of the modules $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$ is carried by the unimodular transformation $[E]$ into the greatest common divisor (respectively, the least common multiple) of the transformed modules; in other words, the identities

$$\begin{aligned} (\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s)E &= (\mathfrak{a}_1E, \mathfrak{a}_2E, \dots, \mathfrak{a}_sE), \\ [\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s]E &= [\mathfrak{a}_1E, \mathfrak{a}_2E, \dots, \mathfrak{a}_sE] \end{aligned}$$

hold.

If A is a square basis of the module $\mathfrak{a}$ and $\mathfrak{a}E = \mathfrak{a}'$, then $\mathfrak{a}' = \text{Md } AE$ and $\text{Cl } \mathfrak{a}' = \text{Cl } AE = \text{Cl } A = \text{Cl } \mathfrak{a}$. Conversely, if $\mathfrak{a}' = \text{Md } A'$ is a module of the class $\text{Cl } \mathfrak{a}$ and A' is square, then by Fundamental Theorem I one may write $A' = E_1AE_2$, and it follows that

$$\mathfrak{a}' = \text{Md } E_1AE_2 = \text{Md } AE_2 = \mathfrak{a}E_2.$$

Thus:

VIII. A module $\mathfrak{a}$ can be carried by unimodular transformations into every module of its class, but into no further module.

This is the basis for the fact that all essential properties of a module are determined by its class.

One of the most important problems to be solved will consist in determining the number of modules which belong to the same class A , given by its system of invariants. We denote this number, which is a “function of A ”, by

$$\psi(A).$$

It is easy to see that for classes whose norm vanishes, $\psi(A) = \infty$, except in the case in which all invariants of A vanish, where evidently one must put $\psi(A) = 1$. We shall therefore be able to restrict ourselves to determining the function ψ for classes of positive norm.

3. Multiplication of modules by a number.

Let $\alpha_1, \alpha_2, \dots, \alpha_r$ be rows of n integers each, and let m be any integer. Then the module $\text{Md}(m\alpha_1, m\alpha_2, \dots, m\alpha_r)$ consists of all rows which arise from the rows of the module $\text{Md}(\alpha_1, \alpha_2, \dots, \alpha_r)$ by multiplication by m . We denote it by “ $m \cdot \mathfrak{a}$ ” and suppose $m > 0$. If $\mathfrak{a}$ is divisible by $\mathfrak{b}$, then $m\mathfrak{a}$ is divisible by $m\mathfrak{b}$, and conversely. If the module $\mathfrak{a}$ has invariants $a_1, a_2, \dots, a_n$, then $m\mathfrak{a}$ has invariants $ma_1, ma_2, \dots, ma_n$; conversely, if a module $\mathfrak{c}$ has invariants $ma_1, ma_2, \dots, ma_n$, one may write $\mathfrak{c} = m\mathfrak{a}'$, and then $a_1, a_2, \dots, a_n$ are the invariants of $\mathfrak{a}'$. Hence, if $A = \text{Cl}(a_1 \mid a_2 \mid \dots \mid a_n)$ is any class and if $\text{Cl}(ma_1 \mid ma_2 \mid \dots \mid ma_n) = mA$ is put, the classes A and mA contain equally many modules, i.e.

$$\text{IX.} \quad \psi(mA) = \psi(A).$$

The class E contains only the module $\mathfrak{e}$, and hence mE contains only the single module $m\mathfrak{e}$, so that for every unimodular system E one has $m\mathfrak{e}E = m\mathfrak{e}$.

X. In order that the module $\mathfrak{a}$ be a divisor or a multiple of $m\mathfrak{e}$, it is necessary (Theorem III) and sufficient that $\text{Cl } \mathfrak{a} = A$ be a divisor or multiple of mE .

Indeed, if the stated condition is fulfilled, the principal module $\mathfrak{a}'$ of A is a divisor or multiple of $m\mathfrak{e}$. By Theorem VIII one can then write $\mathfrak{a} = \mathfrak{a}'E$, whence it follows that $\mathfrak{a}$ is a divisor or multiple of $m\mathfrak{e}E = m\mathfrak{e}$. In entirely similar fashion one proves:

XI. If A is any class of the same degree as E , and $\mathfrak{a}$ is any module of A , then

$$\text{Cl}(\mathfrak{a}, m\mathfrak{e}) = (A, mE), \quad \text{Cl}[\mathfrak{a}, m\mathfrak{e}] = [A, mE].$$

4. Modules of positive norm.

XII. Let $\mathfrak{a}$ be a module of the class A , let $\text{Nm } \mathfrak{a} = \text{Nm } A = a > 0$, let a be the product of the relatively prime numbers a' and a'' , and let A' and A'' be the components of A whose norms are a' and a'' , respectively. Then $\mathfrak{a}$ has one and only one divisor $\mathfrak{a}'$ in the class A' , and likewise one and only one divisor $\mathfrak{a}''$ in the class A'' , and

$$\mathfrak{a} = [\mathfrak{a}', \mathfrak{a}''].$$

XIII. If the classes of the modules $\mathfrak{a}'$ and $\mathfrak{a}''$ are relatively prime, then

$$\text{Cl}[\mathfrak{a}', \mathfrak{a}''] = [\text{Cl } \mathfrak{a}', \text{Cl } \mathfrak{a}''].$$

Proofs. That, under the hypotheses of XII, $\mathfrak{a}$ has a divisor $\mathfrak{a}'$ in the class A' is already asserted by Theorem III. We now show that $\mathfrak{a}$ has no divisor of norm a' other than $\mathfrak{a}'$. Suppose there were a second such module $\mathfrak{a}'_1$. Then $\mathfrak{a}'$ and $\mathfrak{a}'_1$ would be divisors of $\mathfrak{a}$ and, by Theorem X, of $a'\mathfrak{e}$; hence $[\mathfrak{a}', \mathfrak{a}'_1]$ would be a divisor of $(\mathfrak{a}, a'\mathfrak{e})$, $\text{Cl}[\mathfrak{a}', \mathfrak{a}'_1]$ would be a divisor of $\text{Cl}(\mathfrak{a}, a'\mathfrak{e}) = (A, a'E) = A' = \text{Cl } \mathfrak{a}'$, and Theorem IV would imply $[\mathfrak{a}', \mathfrak{a}'_1] = \mathfrak{a}'$, i.e. $\mathfrak{a}'_1$ is a divisor of $\mathfrak{a}'$. Since, however, the norms of $\mathfrak{a}'$ and $\mathfrak{a}'_1$ are positive and equal, it follows that $\mathfrak{a}'_1 = \mathfrak{a}'$. Thus $\mathfrak{a}$ has only one divisor $\mathfrak{a}'$ in A' , and likewise only one divisor $\mathfrak{a}''$ in A'' .

From the fact that the classes $A' = \text{Cl } \mathfrak{a}'$ and $A'' = \text{Cl } \mathfrak{a}''$ are relatively prime, it already follows that $[\mathfrak{a}', \mathfrak{a}'']$ belongs to the class $A = [A', A'']$. Namely, put $\text{Cl}[\mathfrak{a}', \mathfrak{a}''] = A_1$. Then $\text{Nm } A_1 = \text{Nm}[\mathfrak{a}', \mathfrak{a}'']$ is divisible by $[a', a''] = a$, while on the other hand, by Theorem V,

$$\text{Nm } A_1 = \text{Nm}[\mathfrak{a}', \mathfrak{a}''] = \frac{a'a''}{\text{Nm}(\mathfrak{a}', \mathfrak{a}'')} = \frac{a}{\text{Nm}(\mathfrak{a}', \mathfrak{a}'')}$$

is a divisor of a , so $\text{Nm } A_1 = a$. The modules $\mathfrak{a}'$ and $\mathfrak{a}''$, which by the preceding are uniquely determined by the conditions that they have norms a' and a'' and are divisors of $[\mathfrak{a}', \mathfrak{a}'']$, must therefore belong to the components A'_1, A''_1 of A_1 satisfying $\text{Nm } A'_1 = a'$, $\text{Nm } A''_1 = a''$; and these can be identical with A', A'' only if also $A_1 = A$. Finally, since $\mathfrak{a}$ is divisible by $\mathfrak{a}'$ and $\mathfrak{a}''$, and hence by $[\mathfrak{a}', \mathfrak{a}'']$, and since $\text{Cl } \mathfrak{a} = \text{Cl}[\mathfrak{a}', \mathfrak{a}'']$, it follows that

$$\mathfrak{a} = [\mathfrak{a}', \mathfrak{a}''].$$

This proves Theorems XII and XIII.

XIV. Let the module $\mathfrak{a}$, of norm $a(> 0)$, be divisible by the module $\mathfrak{b}$, of norm b . Let a be the product of the relatively prime numbers a' and a'' , and let $\mathfrak{a}', \mathfrak{a}''$ be the divisors of $\mathfrak{a}$ which have norms a' and a'' . Then the modules $\mathfrak{b}' = (\mathfrak{a}', \mathfrak{b})$ and $\mathfrak{b}'' = (\mathfrak{a}'', \mathfrak{b})$ have norms $b' = (a', b)$ and $b'' = (a'', b)$, and their least common multiple is $\mathfrak{b}$.

Proof.

$$\text{Nm}[\mathfrak{a}', \mathfrak{b}] = \frac{a'b}{\text{Nm}(\mathfrak{a}', \mathfrak{b})}$$

is a divisor of $a'b$ and also of $\text{Nm } \mathfrak{a} = a = a'a''$, hence also of $a'(a'', b) = a'b''$. Therefore

$$\text{Nm } \mathfrak{b}' = \text{Nm}(\mathfrak{a}', \mathfrak{b}) = \frac{a'b}{\text{Nm}[\mathfrak{a}', \mathfrak{b}]}$$

is divisible by $a'b/(a'b'') = b/b'' = b'$. Conversely, $\text{Nm } \mathfrak{b}'$ is evidently a divisor of $(a', b) = b'$. Thus $\text{Nm } \mathfrak{b}' = b'$, and similarly $\text{Nm } \mathfrak{b}'' = b''$. Since b' and b'' are relatively prime, it follows from XII and XIII that $\text{Nm}[\mathfrak{b}', \mathfrak{b}''] = [b', b''] = \text{Nm } \mathfrak{b}$, and since $\mathfrak{b}$ is divisible by $[\mathfrak{b}', \mathfrak{b}'']$, it must now be that $[\mathfrak{b}', \mathfrak{b}''] = \mathfrak{b}$.

If $\mathfrak{a}'$ is a divisor of the module $\mathfrak{a}$ and $\text{Cl } \mathfrak{a}' = A'$ is a component of $\text{Cl } \mathfrak{a} = A$, we call $\mathfrak{a}'$ a “component of $\mathfrak{a}$ ”. From XII and XIII it follows that each component of A corresponds to a definite component of $\mathfrak{a}$, and each decomposition of A into two components A', A'' corresponds to a decomposition of $\mathfrak{a}$ into two components $\mathfrak{a}'$ and $\mathfrak{a}''$ such that $[\mathfrak{a}', \mathfrak{a}''] = \mathfrak{a}$. If $\mathfrak{a}'$ is allowed to run through every module of A' and $\mathfrak{a}''$ through every module of A'' , then $[\mathfrak{a}', \mathfrak{a}'']$ represents every module of A exactly once; hence the relation

$$\psi(A) = \psi(A') \cdot \psi(A'') \quad (5)$$

holds. The decomposition of the module $\mathfrak{a}$ can be continued until one reaches its primary components, whose norms are powers of primes. We denote by $\mathfrak{a}_p$ the component of $\mathfrak{a}$ belonging to the class A_p . From XIV one now obtains:

XV. If $\mathfrak{a}$ and $\mathfrak{b}$ are modules of positive norm, then for the divisibility of $\mathfrak{a}$ by $\mathfrak{b}$ it is necessary and sufficient that for every prime p , $\mathfrak{b}_p$ be a divisor of $\mathfrak{a}_p$.

It follows further that:

XVI. If $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$ are modules of positive norm, then the relations

$$(\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s) = \mathfrak{b}, \quad [\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s] = \mathfrak{c}$$

are respectively equivalent to the systems of relations

$$((\mathfrak{a}_1)_p, (\mathfrak{a}_2)_p, \dots, (\mathfrak{a}_s)_p) = \mathfrak{b}_p, \quad [(\mathfrak{a}_1)_p, (\mathfrak{a}_2)_p, \dots, (\mathfrak{a}_s)_p] = \mathfrak{c}_p.$$

Finally, from (5) one also obtains the formula

$$\text{XVII.} \quad \psi(A) = \prod_p \psi(A_p),$$

where the product ranges over all primes, or only over those which divide $\text{Nm } A$, since for the others $A_p = E$ and $\psi(A_p) = 1$.

§ 3. Congruences; the functions φ_r and φ .

Two rows σ, σ' each of n elements are called congruent modulo the module $\mathfrak{a}$ of degree n ,

$$\sigma \equiv \sigma' \pmod{\mathfrak{a}},$$

when $\sigma - \sigma'$ belongs to the module $\mathfrak{a}$. If R, R' are rectangles of degree n with r rows each, then

$$R \equiv R' \pmod{\mathfrak{a}}$$

means that each row of R is congruent to the corresponding row of R' . Each row of n elements represents a definite residue of the module $\mathfrak{a}$, in the sense that congruent rows are regarded as representatives of the same residue. Likewise, a rectangle of degree n with r rows represents a definite r -rowed residue of $\mathfrak{a}$. From the congruence

$$R \equiv R' \pmod{\mathfrak{a}}$$

it follows that

$$RE \equiv R'E \pmod{\mathfrak{a}E},$$

where E denotes any unimodular system. Thus the unimodular transformation $[E]$, which carries $\mathfrak{a}$ into $\mathfrak{a}E$, at the same time carries every residue of $\mathfrak{a}$ into a definite residue of $\mathfrak{a}E$. Since the transformation is uniquely reversible, it establishes a one-to-one correspondence between the residues of $\mathfrak{a}$ and those of $\mathfrak{a}E$. From this and from § 2, VIII it follows that for all modules of a class, the number of residues of a specified row-number is the same. Consideration of the principal module of a class A shows that, if $\text{Nm } A \neq 0$, the number of one-row residues of a module of A is equal to $\text{Nm } A$, and consequently the number of r -rowed residues is equal to $(\text{Nm } A)^r$. If, on the other hand, $\text{Nm } A = 0$, then the number of residues is evidently infinite.

If to a basis of the module $\mathfrak{a}$ one adjoins the rows of a rectangle R , one obtains a basis of the module

$$(\mathfrak{a}, \text{Md } R),$$

which, for brevity, we shall denote simply by

$$(\mathfrak{a}, R).$$

The module $(\mathfrak{a}, R)$ is a divisor of $\mathfrak{a}$. If $R \equiv R' \pmod{\mathfrak{a}}$, then

$$(\mathfrak{a}, R') = (\mathfrak{a}, R).$$

Now suppose that $\text{Nm } \mathfrak{a} > 0$, and that

$$\mathfrak{a} = [\mathfrak{a}', \mathfrak{a}'']$$

is a decomposition of $\mathfrak{a}$ into components, so that $(\mathfrak{a}', \mathfrak{a}'') = \mathfrak{e}$. To every residue R of $\mathfrak{a}$ there then corresponds a definite residue R' of $\mathfrak{a}'$ and a residue R'' of $\mathfrak{a}''$, satisfying the congruences

$$R \equiv R' \pmod{\mathfrak{a}'}, \quad R \equiv R'' \pmod{\mathfrak{a}''}. \quad (1)$$

Conversely, it is easily seen (for example, by comparing the numbers of residues of the modules $\mathfrak{a}, \mathfrak{a}', \mathfrak{a}''$) that to every combination of a residue of $\mathfrak{a}'$ and a residue of $\mathfrak{a}''$ with the same number of rows there belongs a residue R of $\mathfrak{a}$ determined by the congruences (1). We now prove:

I. If

$$\mathfrak{a} = [\mathfrak{a}', \mathfrak{a}'']$$

is a component decomposition of $\mathfrak{a}$, and if the residues R, R', R'' of $\mathfrak{a}, \mathfrak{a}', \mathfrak{a}''$ satisfy the congruences (1), then between the modules

$$\mathfrak{b} = (\mathfrak{a}, R), \quad \mathfrak{b}' = (\mathfrak{a}', R'), \quad \mathfrak{b}'' = (\mathfrak{a}'', R'')$$

there is the equality

$$\mathfrak{b} = [\mathfrak{b}', \mathfrak{b}''].$$

Proof. One has

$$\mathfrak{b}' = (\mathfrak{a}', R') = (\mathfrak{a}', R) = ((\mathfrak{a}, \mathfrak{a}'), R) = (\mathfrak{a}', (\mathfrak{a}, R)) = (\mathfrak{a}', \mathfrak{b}),$$

and similarly $\mathfrak{b}'' = (\mathfrak{a}'', \mathfrak{b})$. Hence by § 2, XIV, $[\mathfrak{b}', \mathfrak{b}''] = \mathfrak{b}$.

Let r be a positive integer, and let the number of r -rowed residues of $\mathfrak{a}$ for which

$$(\mathfrak{a}, R) = \mathfrak{e}$$

be denoted by

$$\varphi_r(\mathfrak{a}).$$

Since for every residue R and every unimodular system E the relation

$$(\mathfrak{a}E, RE) = (\mathfrak{a}, R)E$$

holds, the function φ_r has the same value for all modules of the same class. Thus $\varphi_r(\mathfrak{a})$ is a function only of the class of $\mathfrak{a}$; if $\text{Cl } \mathfrak{a} = A$, we put

$$\varphi_r(\mathfrak{a}) = \varphi_r(A).$$

We now turn to the determination of the function $\varphi_r(A)$, excluding the case $\text{Nm } A = 0$.

Again write

$$\mathfrak{a} = [\mathfrak{a}', \mathfrak{a}'']$$

as a component decomposition of $\mathfrak{a}$. Let R run through every r -rowed residue of $\mathfrak{a}$, and let R', R'' run through every combination of an r -rowed residue of $\mathfrak{a}'$ and one of $\mathfrak{a}''$, subject to the condition that R, R', R'' are connected by congruences (1). Then I shows that the equation

$$(\mathfrak{a}, R) = \mathfrak{e} \tag{2}$$

is equivalent to the system of equations

$$(\mathfrak{a}', R') = \mathfrak{e}, \quad (\mathfrak{a}'', R'') = \mathfrak{e}. \tag{3}$$

Since (2) is satisfied by $\varphi_r(\mathfrak{a})$ residues R , and (3) by $\varphi_r(\mathfrak{a}') \cdot \varphi_r(\mathfrak{a}'')$ combinations R', R'' , it follows that

$$\varphi_r(\mathfrak{a}) = \varphi_r(\mathfrak{a}') \cdot \varphi_r(\mathfrak{a}''),$$

or, putting $\text{Cl } \mathfrak{a}' = A'$ and $\text{Cl } \mathfrak{a}'' = A''$,

$$\varphi_r(A) = \varphi_r(A') \cdot \varphi_r(A'').$$

Continuing the decomposition of A into primary components gives

$$\text{II.} \quad \varphi_r(A) = \prod_p \varphi_r(A_p).$$

Thus it remains only to determine the function φ_r for primary classes.

Let A be the principal system of the primary class $A = A_p$, so $\text{Md } A = \mathfrak{a}$ is its principal module. Suppose that among the invariants of A only the last q are divisible by p , while the first $n - q$ are equal to 1. In order that, for an r -rowed rectangle R , one have $(\mathfrak{a}, R) = \mathfrak{e}$, the determinants of degree n in the rectangle formed from the rows of A and R must have 1 as greatest common divisor. Evidently this is possible only when $r \geq q$. Thus for $r < q$ one has $\varphi_r(A) = 0$. Now suppose $r \geq q$. In order that $(\mathfrak{a}, R) = \mathfrak{e}$, it is necessary and sufficient that the determinants of degree q which can be taken from the last q columns of R should not all be divisible by p . The module $(\mathfrak{a}, p\mathfrak{e})$ is the principal module of the class (A, pE) , whose first $n - q$ invariants are equal to 1 and whose last q invariants are equal to p . If the rectangle R satisfies $(\mathfrak{a}, R) = \mathfrak{e}$, then also $(\mathfrak{a}, p\mathfrak{e}, R) = \mathfrak{e}$; conversely, from $(\mathfrak{a}, p\mathfrak{e}, R) = \mathfrak{e}$ it follows that $(\mathfrak{a}, R) = \mathfrak{e}$. Moreover, under the representatives of a definite r -rowed residue of $(\mathfrak{a}, p\mathfrak{e})$, there are exactly

$$\frac{\text{Nm } \mathfrak{a}}{p^q}$$

residues incongruent modulo $\mathfrak{a}$. Hence one obtains the relation

$$\varphi_r(\mathfrak{a}) = \left(\frac{\text{Nm } \mathfrak{a}}{p^q} \right)^r \cdot \varphi_r(\mathfrak{a}, p\mathfrak{e}). \tag{4}$$

Two rectangles R_1, R_2 are congruent modulo $(\mathfrak{a}, p\mathfrak{e})$ when the rectangles R'_1, R'_2 formed from their last q columns are congruent modulo p . (The statement that two rectangles of the same degree and the same number of rows are congruent modulo a number means that corresponding elements are congruent modulo that number.) Hence $\varphi_r(\mathfrak{a}, p\mathfrak{e})$ is the number of rectangles, incongruent modulo p , with r rows and q ($\leq r$) columns, in which not all determinants of degree q are divisible by p . This number, as a simple calculation gives,³ is

$$p^{qr} \left(1 - \frac{1}{p^r}\right) \left(1 - \frac{1}{p^{r-1}}\right) \cdots \left(1 - \frac{1}{p^{r-q+1}}\right),$$

and therefore, taking (4) into account,

$$\varphi_r(A) = (\text{Nm } A)^r \left(1 - \frac{1}{p^r}\right) \left(1 - \frac{1}{p^{r-1}}\right) \cdots \left(1 - \frac{1}{p^{r-q+1}}\right). \quad (5)$$

This equation remains correct also for $q > r$, since in that case the expression on the right side of (5) vanishes.

Thus we have the following theorem:

III. If $A = A_p$ is a primary class and if q of its invariants are divisible by p (the others therefore being equal to 1), then

$$\varphi_r(A) = (\text{Nm } A)^r \left(1 - \frac{1}{p^r}\right) \left(1 - \frac{1}{p^{r-1}}\right) \cdots \left(1 - \frac{1}{p^{r-q+1}}\right).$$

It is noteworthy that the expression for $\varphi_r(A)$ does not contain the degree of the class A at all. We now introduce another function into the calculation, in which this degree does play a role: if n is the degree of A , then $\varphi_n(A)$ will be denoted simply by $\varphi(A)$. For the function φ one has the relation

$$\text{IIa.} \quad \varphi(A) = \prod_p \varphi(A_p).$$

For abbreviation put

$$\vartheta(p, k) = \left(1 - \frac{1}{p}\right) \left(1 - \frac{1}{p^2}\right) \cdots \left(1 - \frac{1}{p^k}\right) \quad (k > 0),$$

$$\vartheta(p, 0) = 1.$$

Then Theorem III gives:

IIIa. If $A = A_p$ is a primary class of degree n , and if q of its invariants are divisible by p (so the first $n - q$ are equal to 1), then

$$\varphi(A) = (\text{Nm } A)^n \cdot \frac{\vartheta(p, n)}{\vartheta(p, n - q)}.$$

³Compare C. Jordan, *Traité des substitutions*, no. 123.